

SVPM'S COLLEGE OF ENGINEERING, MALEGAON (Bk)
Department of Electronics and Telecommunication Engineering



A Synopsis Report on
Raspberry Pi based Weather Reporting Over IOT

BE E and TC Department

Submitted by

Group No: 8

Mr.Kashid Vishal Mohan Roll No.:222

Mr. Kate Sagar Haridas Roll No.:223

Mr. Lalage Tushar Dasharath Roll No.:231

Under the guidance of Dr.N.B.Dhaigude

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Chapter 1

Project Summary

Title of project:

Raspberry Pi based Weather Reporting Over IOT

Area of Project: IOT

Problem Statement:

- The problem that the real time weather monitoring system using IoT aims to solve is the lack of accurate and timely weather information
- Traditional weather monitoring systems rely on a network of stationary weather stations, which may not provide sufficient coverage and may not be able to provide real-time information. This can lead to inadequate preparation for weather events and potential safety hazards.
- The IoT-based weather monitoring system aims to overcome these limitations by using a network of sensors that are deployed in various locations and can provide real-time data on temperature, humidity, wind speed, and other meteorological parameters

Block Diagram :

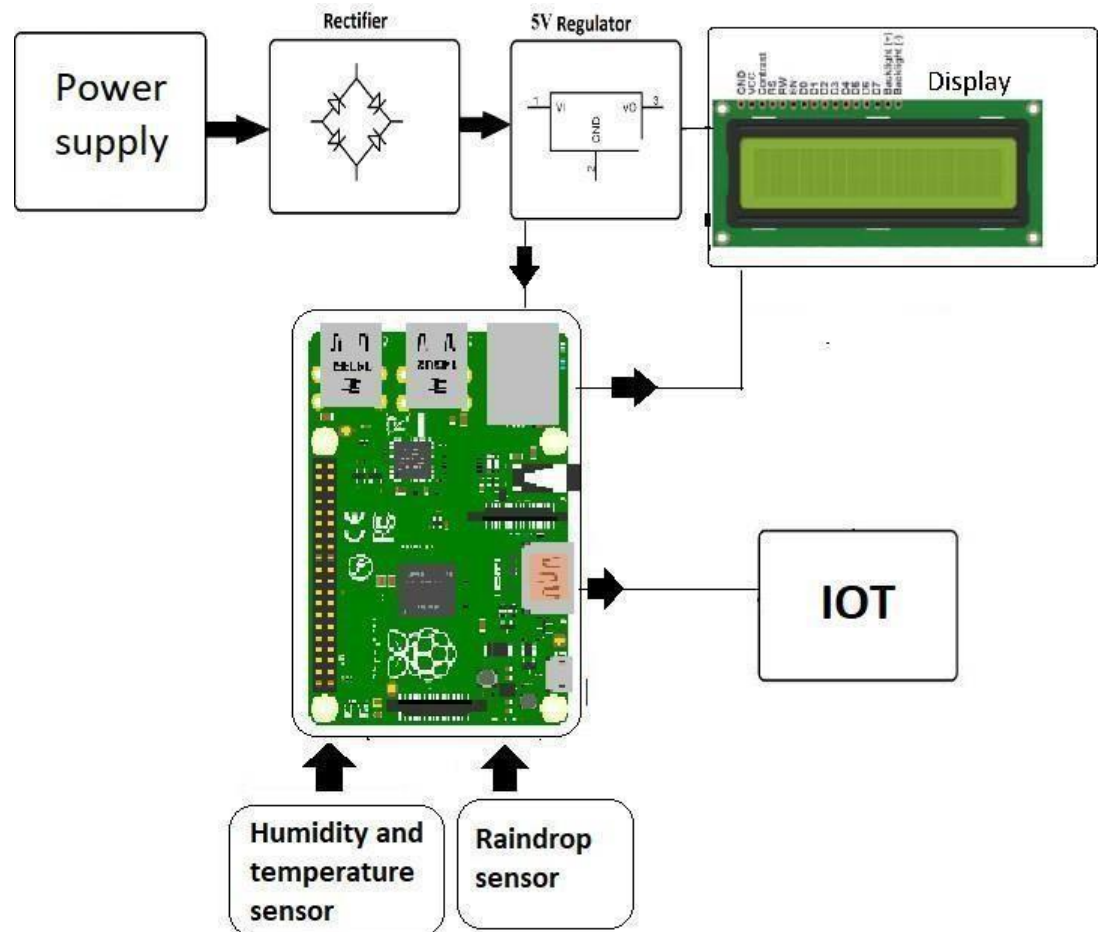


Figure 2.1: Block Diagram for Raspberry Pi based Weather Reporting Over IOT

2. Design Methodology

There are three phases to achieve the stated objectives. Figure 1 shows the architecture of the project.

A. Phase 1-Weather station design and prototype development

The weather parameters are measured using Sparkfun Weather Shield sensor and weather meter connecting to Arduino Uno R3 to capture all the data from the weather sensors. The measurements taken include temperature, humidity, atmospheric pressure, light intensity, wind speed and direction, and rain precipitation.

B. Phase 2-Transferring data to initial state cloud platform

Raspberry Pi 3 Model B is used as a gateway between sensors and IoT platform. It streamed and saved the data to Cloud platform directly via Wi-Fi. Cloud platform that is used in this project is Google Cloud SQL because it fully manages data, supports monitoring, logging, and diagnostics, data analysis and helps developers in building vertical applications.

C. Phase 3-Application of IoT and prediction phase

Stream of weather parameters/information can be displayed in a professional graphical fashion which can be accessed from Google Data Studio webpage. Google also allows the development of web applications that is reliable, fast and engaging. An application is created using Android Studio which allowed users to check the weather using mobile phone. Real time notification will then be sent to the user via social media regularly and irregularly. Weather prediction is done by using normal algorithm (i.e if-else operation) which allows users to get themselves prepared for their future plans.

Chapter 2

Benefits:

Benefits: Using a Raspberry Pi for weather reporting over IoT (Internet of Things) offers a range of benefits:

- 1. Cost-Effective:** Raspberry Pi boards are relatively inexpensive compared to traditional computing systems. This makes them a cost-effective choice for deploying multiple weather stations or sensors.
- 2. Customizability:** The Raspberry Pi's open-source nature allows for extensive customization. You can tailor the hardware and software to meet specific weather monitoring needs, from temperature and humidity to more advanced parameters like wind speed and atmospheric pressure.
- 3. Scalability:** You can easily scale up your weather reporting network by adding more Raspberry Pi units. This flexibility is useful for creating a dense network of weather stations for detailed local or regional monitoring.
- 4. Real-Time Data:** IoT capabilities enable real-time data collection and transmission. With the Raspberry Pi connected to the internet, you can access live weather data from anywhere in the world, facilitating timely decision-making.
- 5. Remote Monitoring and Management:** Once set up, Raspberry Pi-based weather stations can be managed remotely. You can update software, configure settings, and monitor data without needing to physically access each device.
- 6. Integration with Cloud Services:** Raspberry Pi can easily integrate with cloud platforms like AWS, Google Cloud, or Microsoft Azure. This allows for data storage, processing, and advanced analytics, as well as visualization and sharing of weather data.
- 7. Data Logging and Analysis:** The Raspberry Pi can store historical weather data and perform data analysis. This helps in identifying trends and patterns, which can be useful for research or forecasting purposes.

Chapter 3

Expected Output and Applications:

Expected Output:

- Storing weather data into Google Cloud SQL
- Displaying weather data on screen

Applications:

1. Personal Weather Stations

- **Home Monitoring:** Individuals can set up personal weather stations to monitor local weather conditions in real-time. This can include temperature, humidity, and barometric pressure, providing valuable data for gardening, home maintenance, and personal comfort.

2. Smart Agriculture

- **Precision Farming:** Farmers can use Raspberry Pi-based weather stations to monitor soil moisture, temperature, and humidity, optimizing irrigation schedules and improving crop yield

Climate Control: Data from weather stations can be used to automate greenhouse climate controls, such as ventilation and heating systems, based on real-time weather conditions.

- data to larger research initiatives.

3. Smart Cities

- **Urban Weather Networks:** Integrate Raspberry Pi-based weather stations into smart city infrastructure to monitor and analyze urban microclimates, helping manage issues like heat islands and air pollution.
- **Public Information:** Provide real-time weather updates to residents through public displays, apps, or websites, enhancing community awareness and preparedness.

4. Outdoor Activities and Sports

- **Recreational Monitoring:** Provide real-time weather updates for outdoor enthusiasts, such as hikers, cyclists, or boaters, to ensure safety and optimize activities.
- **Sports:** Monitor conditions for outdoor sports, such as wind speed for sailing or temperature for running events.

5. Custom Applications

- **Personal Projects:** Hobbyists and tech enthusiasts can create custom weather reporting solutions tailored to their specific interests or needs, such as monitoring conditions for a hobby farm or a unique environmental setup.

Chapter 4

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Facility Required

Software Required:

1. Python
2. OS: linux

Hardware Required:

- Raspberry Pi 3 –
- DHT11 Digital Temperature Humidity Sensor –
- Rain Drop Sensor -
- LCD's
- Resistors
- Capacitors –
- Transistors –
- Cables & Connectors – Buy Cables & Connectors Online
- Diodes –
- PCB – Buy PCB & Breadboards Online
- LED's – Buy LED Online
- Transformer/Adapter – Buy Transformers & Adapters Online
- Push Button – Buy Buttons & Switches Online

Book Required:

- IEEE Access
- Data Sheets

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Chapter 5

References

Reference Books:

'An Embedded Software Primer' by David E.Simson

'ESP32 Programming for the Internet of Things' by Sever Spanulescu

Free Online Resources:

- Raspberry Pi Documentation and Tutorials: [Arduino.cc](https://www.arduino.cc)
- ESP32 Official Documentation: [Espressif.com](https://www.espressif.com)
- Open-Source Robotics Foundation: [ros.org](https://www.ros.org)
- Instructables and Hackster.io: Websites offering numerous DIY project guides and tutorials.

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Dr.N.B.Dhaigude
Internal Guide

Dr. R. K. Patil
HOD

Dr. S. M. Mukane
Principal